

Name
Date
Course

Applications (Homework)

Solve the following equations. Write the Roman numeral of each exponent rule that you use in the margin. Round to the nearest ten thousandth if necessary.

$$\textcircled{1} e^{4x} \cdot e^{2x} = 3$$

$$\textcircled{2} 5 = \frac{10^{3x}}{10^{5x}}$$

$$\textcircled{3} 6 + 4e^x = 8$$

$$\textcircled{4} 14 = 2(10^x) - 1$$

Solve the following equations. Write the Roman numeral of each exponent rule that you use in the Margin.
Round to the nearest hundredth if necessary.

⑤ $7 = 3 \log x - 1$

⑥ $\frac{\ln x}{2} - 16 = -14$

Given the following intensities of sound, find the loudness perceived by the human ear.

⑦ 10^{-7} W/m^2

⑧ 10^{-2} W/m^2

⑨ 10^{-4} W/m^2

Find the intensity of sound required to produce the following number of decibels.

⑩ 170 dB

⑪ 80 dB

⑫ 49 dB

⑬ If \$21,340 is deposited in a bank account paying a 2% annual interest rate compounded annually, how much money will be in the account in 10 years? Assume that there are no withdrawals or additional deposits

⑭ If \$1,090 is deposited in a bank account paying a 1.7% annual interest rate compounded semiannually, how much money will be in the account in 4 years? Assume that there are no withdrawals or additional deposits.

⑮ If interest is compounded continuously for 6 years at a 1.8% rate, how much would a \$3,500 investment grow to?

⑯ If a 4% interest rate is compounded continuously for 5 years, and the initial investment is \$600, how much would the investment be worth?

⑰ If a \$20,000 investment grows to \$25,334 in 7 years, what interest rate was applied if it was compounded continuously?

⑮ If an investment grows from \$819 to \$1200 at a 14% annual interest rate compounded continuously, how much time passed during this growth?

Use Logarithm Property IX to change each logarithm to a quotient of logarithms with the indicated base.

⑰ Base 7.

$$\log_8 5$$

⑲ Base 40.

$$\log_2 7$$

⑱ Base 5.

$$\log_4 9$$

⑳ Base 3.

$$\log_4 10$$

Use your calculator's base 10 logarithm button (LOG) and Logarithm Property IX to find the value of each logarithm below. Round to the nearest thousandth.

㉓ $\log_7 8$

㉔ $\log_2 5$

㉕ $\log_4 9$

Use your calculator's base "e" logarithm button (LN) and Logarithm Property IX to find the value of each logarithm below. Round to the nearest thousandth.

㉖ $\log_9 10$

㉗ $\log_{15} 8$

②⑧ $\log_3 19$

Answers

① $x \approx 0.1831$

② $x \approx -0.3495$

③ $x \approx -0.6931$

④ $x \approx 0.8751$

⑤ $x \approx 464.16$

⑥ $x \approx 54.6$

⑦ 50 dB

⑧ 100 dB

⑨ 80 dB

⑩ $100,000 \text{ W/m}^2$ or 10^5 W/m^2

⑪ 10^{-4} W/m^2 or 0.0001 W/m^2

⑫ $10^{-7.1} \text{ W/m}^2$

⑬ $\$26,013.34$

⑭ $\$1,166.36$

⑮ $\$3,899.17$

⑯ $\$732.84$

⑰ 3.38%

⑱ 2.7285 years

⑲ $\frac{\log_7 5}{\log_7 8}$

⑳ $\frac{\log_5 9}{\log_5 4}$

㉑ $\frac{\log_{40} 7}{\log_{40} 2}$

㉒ $\frac{\log_3 10}{\log_3 4}$

㉓ $\frac{\log 8}{\log 7} \approx 1.069$

㉔ $\frac{\log 5}{\log 2} \approx 2.322$

㉕ $\frac{\log 9}{\log 4} \approx 1.585$

㉖ $\frac{\ln 10}{\ln 9} \approx 1.048$

㉗ $\frac{\ln 8}{\ln 5} \approx 0.768$

㉘ $\frac{\ln 19}{\ln 3} \approx 2.68$